Entropy-Based Causal Analysis of Precursors to Runway Excursions

Rachel Newman

Spring 2026

1 Purpose of This Document

This document defines the *core scientific idea*, scope, and methodological expectations for the capstone project. It is intended as both a starting point and preparation guide, not as a template for final results.

The goal of this project is to investigate how *information-theoretic measures of causality* can be used to identify **precursor variables that causally influence runway excursions**, using only publicly available data.

2 Motivation

According to the International Air Transport Association (IATA), Runway excursions (RE) remain one of the main contributing categories of aviation safety events, accounting for 21 percent of the aviation industry's total accidents over the last ten years (2015 - 2024) (Figure 1). [1]. A runway excursion is an aviation safety event in which an aircraft departs the runway in use during takeoff or landing.

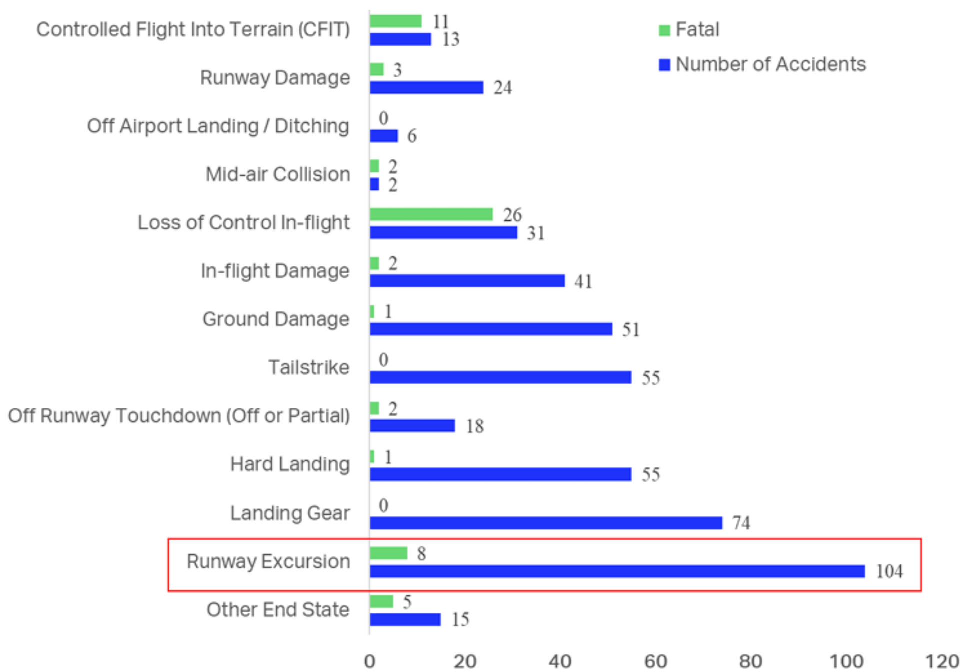

Figure 1: Runway excursions compared to other accident end states over the last decade, adapted from IATA Runway Excursion Risk Assessment (2025).

While RE are often attributed to individual factors such as weather conditions, runway contamination, or pilot technique, incident investigations and safety studies consistently suggest that these events arise from complex interactions among environmental, operational, and human factors.

Traditional aviation safety analyses typically rely on linear statistical models, expert-defined causal chains, or post-incident qualitative assessments. Although these approaches are valuable, these approaches face limitations when multiple interacting variables are present, dependencies are nonlinear, safety events are rare but critical, or correlations are mistaken for causes.

This project proposes an alternative perspective: *causality as information flow*, quantified using entropy-based measures that explicitly condition on confounding variables.

3 Core Scientific Question

The central scientific question guiding this project is:

Which observable environmental, runway, flight, and operational variables provide *unique causal information* about the occurrence of runway excursions, after conditioning on other relevant variables?

The emphasis of this question is on **causal relevance**, not prediction accuracy.

4 Conceptual Framework

4.1 Information-Theoretic View of Causality

Let X, Y, and Z denote random variables derived from observed time series data.

- Mutual Information $I(X; Y)$ quantifies statistical dependence.
- Conditional Mutual Information $I(X; Y|Z)$ quantifies dependence *after* accounting for Z.

A variable X is considered *causally relevant* to Y if

$$I(X; Y|Z) > 0$$

where Z includes all other relevant variables that could mediate or confound the relationship. This principle underlies causation entropy and related causal discovery methods.

4.2 Why Conditioning Matters

Without conditioning:

- correlated variables may appear important,
- indirect effects may be mistaken for direct causes,
- operational or environment confounders may dominate results.

The project explicitly tests whether apparent predictors retain informational value *after conditioning*.

5 Data Sources

The project solely relies on publicly available data. Core sources include:

- NTSB incident and accident databases.
- ICAO E-library of Final Reports: incident reports prior to 2020 sent to ICAO.
- FAA Accident and Incident Data System (AIDS).
- FAA operations and airport data.
- NOAA Aviation Weather Data: METAR and TAF reports.
- NASA Aviation Safety Reporting System (ASRS): incident reports and metadata (primarily used for validation).

6 Event-Centered Analysis

Rather than modeling long continuous histories, the project adopts an *event-centered approach.* Runway excursions are treated as discrete safety events, and the analysis focuses on conditions leading up to each event:

- Identification of runway excursions,
- Define a pre-event time window,
- Extract multivariate time series leading up to each event.

This framing emphasizes **precursors**, not outcomes.

7 Methodological Expectations

This project will be implemented in three stages: entropy estimation, causal screening, and causal graph construction.

7.1 Entropy Estimation

- Estimate entropy and mutal information from finite observational samples.
- Explicitly address challenges related to data sparsity, rare events, and outliers commonly found in aviation safety events.
- Justify the choice of entropy and mutual information estimators, along with any preprocessing or discretization assumptions.

7.2 Causal Screening

- Identify variables with nonzero mutual information with runway excursions.
- Distinguish informative variables from causal ones via conditioning.

7.3 Causal Graph Construction

- Construct directed graphs representing inferred causal structure.
- Compare across different operational contexts or subsets of runway excursion events to evaluate changes in causal relationships.

8 Interpretation and Safety Relevance

A central goal of this project is interpretation rather than optimization. The results will be analyzed to address the following:

- which variables act as early warning indicators for runway excursions,
- which variables are predictive but not causally relevant,
- how causal structure changes under adverse conditions (e.g., degraded runway surfaces or poor weather).

Causal claims will be framed as observational and conditional, not interventional.

9 Explicit Non-Goals

This project explicitly does **not** aim to:

- build black-box predictive models,
- optimize classification performance,
- claim counterfactual control without intervention,
- replace expert judgment or operational knowledge.

Limitations due to finite sample sizes, reporting bias, and observational data constraints will be acknowledged throughout the analysis.

10 Expected Outcomes

The project is expected to demonstrate how information-theoretic tools can be used to identify and interpret causally relevant precursor variables in aviation safety events, contributing to improved understanding of runway excursion risk.

11 Final Remark

This project treats runway excursions as a *scientific inference problem* rather than a prediction task, emphasizing rigor, transparency, and interpretability in causal reasoning.

References

[1] International Air Transport Association (IATA). Runway excursion risk assessment, 2025. Version 1.0 July 2025.